Notes

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1 Failure on high-dimensional circular data

We have shown that flow matching with path integration works well for high-dimensional non-circular data; here we examine whether it works for *circular* tuning on θ using the toy family below.

1.1 Toy dataset

We use a synthetic conditional model for a scalar stimulus θ and a multivariate observation $x \in \mathbb{R}^d$ with $d \geq 2$. Stimuli are drawn independently from a uniform density on a bounded interval,

$$\theta \sim \mathrm{Uniform}(\theta_{\mathrm{low}}, \theta_{\mathrm{high}}). \tag{1}$$

Conditional mean (tuning curve). For each coordinate $j \in \{1, \ldots, d\}$ fix a phase $\phi_j = 2\pi(j-1)/d$. Given θ, the mean of x_j is a cosine in θ,

$$\mu_j(\theta) = A\cos(\omega\theta + \phi_j), \qquad j = 1, \ldots, d, \tag{2}$$

with amplitude $A > 0$ and frequency $\omega > 0$ on the stimulus axis (in the default bundle, $A = \omega = 1$). The conditional mean vector is $\mu(\theta) = (\mu_1(\theta), \ldots, \mu_d(\theta))^\mathsf{T}$.

Observation noise. Conditionally on θ, the observation is multivariate Gaussian with diagonal covariance. Let $\sigma_{\mathrm{base},j} > 0$ be nonnegative baseline scales across neurons; in the usual configuration they interpolate linearly in j between fixed σ_1 and σ_2, and often $\sigma_1 = \sigma_2 = \sigma$ so all coordinates share the same σ. Let α be a fixed activity–variance coupling strength (the arithmetic mean of two nonnegative hyperparameters in the implementation), and let $\varepsilon > 0$ be a negligible variance floor. The per-neuron conditional variances are

$$v_j(\theta) = d\,\sigma_{\mathrm{base},j}^2\left(1 + \alpha\,|\mu_j(\theta)|\right) + \varepsilon, \qquad j = 1, \ldots, d, \tag{3}$$

with independence across neurons,

$$x \mid \theta \sim \mathcal{N}\big(\mu(\theta),\, \mathrm{diag}(v_1(\theta), \ldots, v_d(\theta))\big). \tag{4}$$

Thus noise magnitude increases with $|\mu_j(\theta)|$, and each neuron's noise standard deviation scales proportionally to $\sqrt{d}$ for fixed $\sigma_{\mathrm{base},j}$, α, and $\mu_j(\theta)$. The leading factor d matches the "$\sqrt{d}$ noise scaling" construction used elsewhere in these notes: as d grows, per-coordinate variance grows with d so that dimension alone does not drive the observation toward an effectively noiseless, trivially informative regime for θ.

1.2 Flow matching with path integration failed on 3D circular dataset

Conditional flow matching trains a stimulus-conditional velocity by regressing on the affine bridge law in (**??**)–(**??**). The Fisher-score surrogate uses the Gaussian probability path (**??**) and the same velocity-to-score map (**??**) as in that note. Figure 1 shows the outcome on the $d = 3$ circular toy above. The Hellinger matrix (top row) stays far from the approximate ground truth: off-diagonal Spearman correlation with that reference barely increases with training size n. The decoding matrix (bottom row) does better but still does not match the reference at the largest n shown.

This behavior is expected in hindsight: turning a learned velocity into a score and integrating along θ is an indirect route when the stimulus acts through periodic tuning, so approximation error can accumulate.

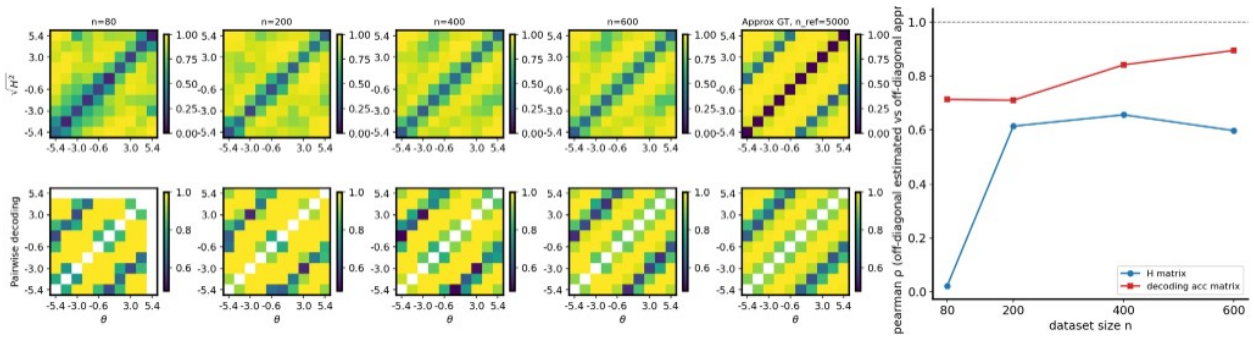

Figure 1: Flow matching with path integration on the $d = 3$ circular (cosine Gaussian $\sqrt{d}$) toy. Top row: estimated $\sqrt{H^2}$ RDMs; bottom row: pairwise decoding matrices; columns are training sizes n and an approximate ground truth from a large reference draw. Right: Spearman correlation between off-diagonal entries of each estimated matrix and the corresponding approximate ground truth, as a function of n (Hellinger in blue, decoding in red).

1.3 Failure on high-dimensional circular data using exact likelihood computation with flow matching on θ space and x space

Flow matching also supports a more direct route: integrate the divergence of the learned velocity along an ODE to obtain a log-density, without converting to a score or integrating a score along θ. Let v_ϕ denote the learned velocity field on observation space. Then the density carried along the flow obeys

$$\frac{\mathrm{d}}{\mathrm{d}t} \log p_t(x_t \mid \theta) = -\nabla_{x_t} \cdot v_\phi(x_t, t \mid \theta) \quad \text{for } x_t \in \mathbb{R}^d, \tag{5}$$

and integrating in time gives the endpoint log-likelihood

$$\log p(x_1 \mid \theta) = \log p_0(x_0) - \int_0^1 \big(\nabla_{x_t} \cdot v_\phi\big)(x_t, t \mid \theta)\, \mathrm{d}t, \tag{6}$$

along the trajectory of $\frac{\mathrm{d}}{\mathrm{d}t} x_t = v_\phi(x_t, t \mid \theta)$, with $\log p_0$ a standard Gaussian on $\mathbb{R}^d$ at $t = 0$. One may evaluate $\log p(x \mid \theta)$ either by the x-space construction in the next paragraph ("x-likelihood"), or by the θ-field construction after it ("θ-likelihood").

x-likelihood. Here the setup is the most direct: fit *one* conditional flow-matching model on observation space with $\frac{\mathrm{d}}{\mathrm{d}t} x_t = v_\phi(x_t, t \mid \theta)$, where θ is fixed when training and at evaluation time. For each stimulus value θ_j and each observation x_i, run the ODE likelihood in (5)–(6) with that θ_j to obtain $\log p(x_i \mid \theta_j)$. There is no separate prior flow on θ and no Bayes step: the matrix with entries $C_{ij} = \log p(x_i \mid \theta_j)$ already targets the conditional log-likelihoods needed downstream (with the usual ΔL diagonal centering applied in the same way as for other C matrices in this project).

θ-likelihood. The idea is to run the same likelihood formula on the *stimulus* θ instead of on x, using two separate flow fits. The first model ignores the observation and follows $\frac{\mathrm{d}}{\mathrm{d}t} \theta_t = v_{\text{prior}}(\theta_t, t)$; it supplies the prior density $\log p(\theta)$. The second model conditions on a fixed x and follows $\frac{\mathrm{d}}{\mathrm{d}t} \theta_t = v_{\text{post}}(\theta_t, t \mid x)$; it supplies $\log p(\theta \mid x)$. In both cases the endpoint log-density is

$$\log p(\theta_1) = \log p_0(\theta_0) - \int_0^1 \frac{\partial v}{\partial \theta}(\theta_t, t)\, \mathrm{d}t, \tag{7}$$

where θ_t solves the ODE for the active field, v is either v_{prior} or $v_{\text{post}}(\cdot \mid x)$, and $\log p_0(\theta_0) = -\frac{1}{2}\big(\theta_0^2 + \log(2\pi)\big)$ for a standard normal base on $\mathbb{R}$. Subtracting the prior evaluation from the posterior evaluation at the same θ yields $\log p(\theta \mid x) - \log p(\theta)$, which Bayes' rule rewrites as $\log p(x \mid \theta) - \log p(x)$. For a fixed observation x_i, that missing term is the same for every candidate θ_j in a row of the pairwise matrix $C_{ij} = \log p(\theta_j \mid x_i) - \log p(\theta_j)$. The subsequent ΔL step removes such row-wise constants, so the contrasts line up with differences in $\log p(x_i \mid \theta_j)$.

Fourier features for θ in the velocity networks. Besides choosing whether likelihoods are accumulated in x-space or in θ-space, one can change how θ is presented to the velocity network. The default is to concatenate the raw scalar θ (together with time and, depending on the field, x or other inputs) as inputs to a standard multilayer architecture. An alternative is a *Fourier-feature* parameterization: map θ to a vector that may include a constant term, the identity term θ, and, for harmonics $k = 1, \ldots, K$ and a fixed angular scale $\omega > 0$, the pairs $\sin(k\omega\theta)$ and $\cos(k\omega\theta)$, then pass that vector through the same style of network. This makes periodic structure in θ easier to represent and is available for both posterior and prior θ-flows, and analogously for the conditional x-flow where θ enters only as a conditioning variable; the harmonic count K and ω are part of the experimental design alongside architecture width and depth.

1.4 Failure of methods on high-dimensional circular data: results

Figure 2. I tried also using Spearman correlation, does work.

In fact, even increasing the number of data the flow matching method still does not work well, I need to check it.

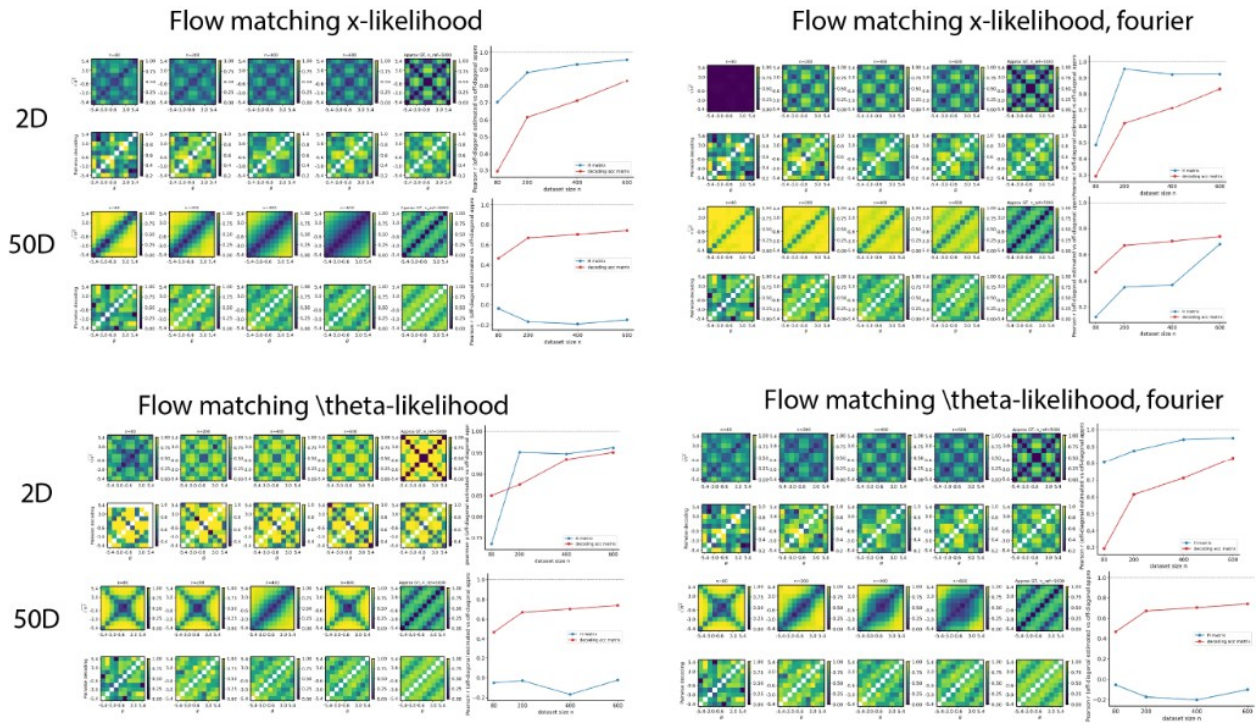

Figure 2: Flow-matching likelihoods: x-likelihood (top left), x-likelihood with Fourier features on θ (top right), θ-likelihood (bottom left), θ-likelihood with Fourier features (bottom right). Within each panel, $d = 2$ then $d = 50$; for each d, top row $\sqrt{\Sigma^{-1}}$–style matrix, bottom row Pearson decoding matrices, columns $n \in \{80, 200, 400, 600\}$ and an approximate ground truth ($n_{\text{ref}} = 5000$). Side plots: Pearson r (diagonal of estimate versus off-diagonal match to the reference) versus n for the H/precision matrix (blue) and the decoding-accuracy matrix (red).

2 A pair-conditioned bridge extension of CTSM-v for continuous parameterized densities

2.1 Problem setup

We consider a continuous family of conditional densities

$$p(x \mid \theta), \qquad \theta \in \Theta \subseteq \mathbb{R}, \tag{8}$$

and a dataset of paired observations

$$\mathcal{D} = \{(\theta_n, x_n)\}_{n=1}^{N}. \tag{9}$$

Our goal is to estimate, for any pair (i, j),

$$r_{ij}(x_i) := \log \frac{p(x_i \mid \theta_i)}{p(x_i \mid \theta_j)}. \tag{10}$$

More generally, for any x and any pair of endpoint parameters (a, b), we would like to estimate

$$r(x; a, b) := \log \frac{p(x \mid b)}{p(x \mid a)}. \tag{11}$$

The guiding idea is to extend the two-sample CTSM-v framework from the case of two unknown endpoint densities to the case of a continuously indexed family $p(x \mid \theta)$. The extension is based on learning a *pair-conditioned bridge model* between the endpoint conditionals $p(x \mid a)$ and $p(x \mid b)$.

2.2 Bridge representation of the conditional log-ratio

For any ordered pair (a, b), define a probability path

$$q_t^{a,b}(x), \qquad t \in [0, 1], \tag{12}$$

with endpoint constraints

$$q_0^{a,b}(x) = p(x \mid a), \qquad q_1^{a,b}(x) = p(x \mid b). \tag{13}$$

Let

$$s_t^{a,b}(x) := \partial_t \log q_t^{a,b}(x) \tag{14}$$

be the corresponding time score. Then, by the fundamental theorem of calculus,

$$\log \frac{p(x \mid b)}{p(x \mid a)} = \log q_1^{a,b}(x) - \log q_0^{a,b}(x) = \int_0^1 s_t^{a,b}(x)\, dt. \tag{15}$$

Therefore, if we can learn the time score $s_t^{a,b}(x)$ for each endpoint pair (a, b), then the desired conditional log-ratio is obtained by integrating over t.

2.3 Pair-conditioned Gaussian bridge

A natural extension of the two-sided CTSM-v bridge is the following conditional Gaussian bridge. Given endpoint samples

$$x_0 \sim p(\cdot \mid a), \qquad x_1 \sim p(\cdot \mid b), \qquad \varepsilon \sim \mathcal{N}(0, I), \tag{16}$$

we define

$$x_t = (1-t)x_0 + tx_1 + \sqrt{\sigma^2 t(1-t)}\,\varepsilon, \qquad t \in [0,1]. \tag{17}$$

Conditioned on (x_0, x_1), the bridge distribution is Gaussian:

$$q_t(x \mid x_0, x_1, a, b) = \mathcal{N}\big(x; (1-t)x_0 + tx_1, \sigma^2 t(1-t)I\big)\,. \tag{18}$$

Marginalizing over the endpoint samples yields the unconditional bridge marginal

$$q_t^{a,b}(x) = \iint q_t(x \mid x_0, x_1, a, b)\, p(x_0 \mid a)\, p(x_1 \mid b)\, dx_0\, dx_1. \tag{19}$$

By construction,

$$q_0^{a,b}(x) = p(x \mid a), \qquad q_1^{a,b}(x) = p(x \mid b), \tag{20}$$

so (15) holds for the desired pairwise ratio.

2.4 Neural parameterization

We learn a vector-valued model

$$f_\phi(x, t, m, \Delta) \in \mathbb{R}^d, \tag{21}$$

where $d = \dim(x)$ and where the pair of endpoint parameters (a, b) is reparameterized as

$$m = \frac{a+b}{2}, \qquad \Delta = b - a. \tag{22}$$

Thus, the model input is

$$(x, t, m, \Delta), \tag{23}$$

rather than (x, t, a, b). This parameterization separates the pair into a midpoint location m and a relative displacement Δ, which is often a more natural coordinate system for learning pair-conditioned transformations.

The scalar time score estimate is obtained by summing the vector output,

$$\hat{s}_\phi(x, t, m, \Delta) := \mathbf{1}^\top f_\phi(x, t, m, \Delta), \tag{24}$$

which matches the CTSM-v style in which a vector-valued regression target is learned and then reduced to the scalar time score.

2.5 CTSM-v training objective

The key advantage of the bridge (17) is that conditioned on (x_0, x_1), the path is Gaussian, so the CTSM-v target remains available in closed form.

Let the bridge variance be denoted by

$$\mathrm{Var}(x_t \mid x_0, x_1) = \sigma^2 t(1-t)I. \tag{25}$$

Introduce the shorthand

$$\mu_t = (1-t)x_0 + tx_1, \qquad x_t = \mu_t + \sqrt{\sigma^2 t(1-t)}\,\varepsilon. \tag{26}$$

For the two-sided bridge, the vector CTSM-v target takes the form

$$g(x_t, \varepsilon, x_0, x_1, t) = -\frac{\tau_t}{\sqrt{2}} + \frac{\tau_t}{\sqrt{2}}\,\varepsilon^{\odot 2} + \frac{\sqrt{2}\,\sqrt{t(1-t)}}{\kappa_t \sigma}\,\varepsilon \odot (x_1 - x_0), \tag{27}$$

where

$$\kappa_t = \sqrt{1 - 4t + 4t^2 + 2\alpha t - 2\alpha t^2}, \qquad \tau_t = \frac{1-2t}{\kappa_t}, \tag{28}$$

and α is the time-weighting factor used in CTSM-v. The corresponding scalar prefactor is

$$\lambda_t = \frac{\sqrt{2}\,t(1-t)}{\kappa_t}. \tag{29}$$

Here $\varepsilon^{\odot 2}$ denotes elementwise squaring and $\odot$ denotes elementwise multiplication.

The CTSM-v loss is then

$$\mathcal{L}(\phi) = \mathbb{E}\left[\|g(x_t, \varepsilon, x_0, x_1, t) - \lambda_t f_\phi(x_t, t, m, \Delta)\|_2^2\right], \tag{30}$$

where the expectation is over the training sampling procedure:

$$a, b \sim \pi(a, b), \qquad x_0 \sim p(\cdot \mid a), \qquad x_1 \sim p(\cdot \mid b), \qquad t \sim \mathrm{Uniform}(\varepsilon_0, 1-\varepsilon_0), \qquad \varepsilon \sim \mathcal{N}(0, I). \tag{31}$$

The pair-conditioning variables are then set to

$$m = \frac{a+b}{2}, \qquad \Delta = b - a. \tag{32}$$

In a dataset-based implementation, one may form training tuples by selecting two observed pairs

$$(\theta_i, x_i), \qquad (\theta_j, x_j), \tag{33}$$

and identifying

$$a = \theta_i, \qquad b = \theta_j, \qquad x_0 = x_i, \qquad x_1 = x_j. \tag{34}$$

This produces a pair-conditioned bridge sample directly from observed data.

2.6 Inference

Once the network has been trained, the estimated scalar time score is

$$\hat{s}_\phi(x, t, m, \Delta) = \mathbf{1}^\top f_\phi(x, t, m, \Delta). \tag{35}$$

For a query point x and endpoint parameters (a, b), define

$$\hat{r}_\phi(x; a, b) := \int_0^1 \hat{s}_\phi\left(x, t, \frac{a+b}{2}, b-a\right) dt. \tag{36}$$

Then the desired pairwise estimator is

$$\log \widehat{\frac{p(x_i \mid \theta_i)}{p(x_i \mid \theta_j)}} = \hat{r}_\phi(x_i; \theta_j, \theta_i). \tag{37}$$

The order of the endpoints matters: the denominator parameter is the start point and the numerator parameter is the end point.

2.7 Statistical assumption and interpretation

This construction is meaningful only under a regularity assumption on the family $p(x \mid \theta)$. In particular, the method relies on the idea that the conditional density varies sufficiently smoothly with θ so that information can be shared across different endpoint pairs through the parameterized model $f_\phi(x, t, m, \Delta)$.

If the data contain very sparse observations at each θ, then the quality of the learned ratio field depends critically on the extent to which the conditional family is learnable from shared structure across nearby and distant parameter values. Under this interpretation, the pair-conditioned bridge model is best viewed as a nonparametric or semiparametric estimator of the full conditional ratio field

$$(x, a, b) \mapsto \log \frac{p(x \mid b)}{p(x \mid a)}, \tag{38}$$

represented through the integral of a learned time score.

2.8 Summary

The proposed extension replaces the original two-distribution CTSM-v setting with a family of pair-conditioned bridges indexed by endpoint parameters (a, b), or equivalently by midpoint–displacement coordinates (m, Δ). The main ingredients are:

1. For each endpoint pair (a, b), define the two-sided Gaussian bridge

$$x_t = (1-t)x_0 + tx_1 + \sqrt{\sigma^2 t(1-t)}\,\varepsilon, \qquad x_0 \sim p(\cdot \mid a), \quad x_1 \sim p(\cdot \mid b).$$

2. Learn a vector-valued CTSM-v network

$$f_\phi(x, t, m, \Delta), \qquad m = \frac{a+b}{2}, \quad \Delta = b - a,$$

   using the closed-form conditional Gaussian target.

3. Recover the conditional log-ratio by integration:

$$\log \frac{p(x \mid b)}{p(x \mid a)} \approx \int_0^1 \mathbf{1}^\top f_\phi\left(x, t, \frac{a+b}{2}, b-a\right) dt.$$

This yields a direct CTSM-v-style estimator for the desired pairwise quantities

$$\log \frac{p(x_i \mid \theta_i)}{p(x_i \mid \theta_j)}.$$

2.9 Hellinger distance and the log-likelihood ratio

Denoting the continuous-valued stimulus variable as θ, the neural population response is modeled as a conditional distribution $p(x \mid \theta)$ which is unknown. A dataset is sampled from the joint distribution

$$p(x, \theta) = p(\theta)p(x \mid \theta), \tag{39}$$

where $p(\theta)$ is the prior over stimulus conditions.

Representation dissimilarity analysis measures the "distance" between two stimulus conditions through the distance between the corresponding conditional distributions. A common choice is

Kullback–Leibler divergence; however, KL divergence is not a metric. We therefore use the squared Hellinger distance $H^2(p_i, p_j)$,

$$H^2(p_i, p_j) = 1 - \int \sqrt{p(x \mid \theta_i)\, p(x \mid \theta_j)}\, dx, \qquad p_i := p(x \mid \theta_i), \;\; p_j := p(x \mid \theta_j). \tag{40}$$

Besides being a true distance, the Hellinger distance is attractive here for two additional reasons: locally (for infinitesimally close conditions) H^2 matches the Fisher-information metric to leading order, and it controls the Bayes error of the optimal binary classifier between p_i and p_j.

Define the log-likelihood ratio

$$\Delta\ell_{ij}(x) := \log p(x \mid \theta_j) - \log p(x \mid \theta_i). \tag{41}$$

Then the Bhattacharyya integrand can be written as

$$\sqrt{p(x \mid \theta_i)\, p(x \mid \theta_j)} = p(x \mid \theta_i) \exp\left(\frac{\Delta\ell_{ij}(x)}{2}\right), \tag{42}$$

which yields the exact one-sided identity

$$H^2(p_i, p_j) = 1 - \mathbb{E}_{x \sim p_i}\left[\exp\left(\frac{\Delta\ell_{ij}(x)}{2}\right)\right]. \tag{43}$$

Although exact, this form is asymmetric at the sample level because the expectation is taken only under p_i. To obtain a symmetric sample-wise quantity, we use

$$H^2(p_i, p_j) = \frac{1}{2}\, \mathbb{E}_{x_i \sim p_i,\, x_j \sim p_j}\left[h_{ij}(x_i) + h_{ij}(x_j)\right], \tag{44}$$

where

$$h_{ij}(x) := 1 - \operatorname{sech}\left(\frac{\Delta\ell_{ij}(x)}{2}\right). \tag{45}$$

This expresses H^2 as an average of two contributions, one evaluated on $x_i \sim p_i$ and one on $x_j \sim p_j$.

The remaining task is to estimate $\Delta\ell_{ij}(x)$ without discretizing θ. In an earlier score-based pipeline, $\Delta\ell_{ij}$ was obtained by integrating an estimated Fisher score along θ. Here we estimate the log-likelihood ratio *directly* from the pair-conditioned bridge time score: by (15),

$$\Delta\ell_{ij}(x) = \int_0^1 s_t^{\theta_i, \theta_j}(x)\, dt, \tag{46}$$

and subsection 2.6 replaces s_t by the learned $\hat{s}_\phi$ from (24).

2.10 From CTSM-v estimates to the Hellinger matrix

For each ordered pair (i, j) in the dataset, set

$$m_{ij} := \frac{\theta_i + \theta_j}{2}, \qquad \Delta_{ij} := \theta_j - \theta_i. \tag{47}$$

Following (36), define the *directed* log-ratio estimate evaluated on observation x_i by

$$\Delta L_{ij} := \int_0^1 \hat{s}_\phi(x_i, t, m_{ij}, \Delta_{ij})\, dt \;\approx\; \log \frac{p(x_i \mid \theta_j)}{p(x_i \mid \theta_i)}. \tag{48}$$

Thus $\Delta L_{ii} = 0$, and ΔL_{ij} is exactly the quantity needed inside h_{ij} after training the pair-conditioned CTSM-v model (30).

The directed sample-wise contribution is

$$H_{ij}^{\rightarrow} := 1 - \operatorname{sech}\left(\frac{\Delta L_{ij}}{2}\right), \tag{49}$$

and its symmetrization is

$$H_{ij}^{\mathrm{sym}} := \frac{1}{2}\left(H_{ij}^{\rightarrow} + H_{ji}^{\rightarrow}\right). \tag{50}$$

Finally, the squared Hellinger distance matrix is obtained by taking expectations over independent draws from the two conditions,

$$\mathbf{H}^2 := \left[\mathbb{E}_{x_i \sim p_i,\, x_j \sim p_j}[H_{ij}^{\mathrm{sym}}]\right], \tag{51}$$

which in practice may be estimated by binning θ or by regression. The role of pair-conditioned CTSM-v is confined to the log-ratio step (48); the Hellinger construction above is unchanged.

Relation to (37)**.** The estimator (37) targets $\log p(x_i \mid \theta_i) - \log p(x_i \mid \theta_j)$, i.e. the negative of $\Delta\ell_{ij}(x_i)$. Consistency requires $\Delta L_{ij} = -\,\hat{r}_\phi(x_i; \theta_j, \theta_i)$ with $\hat{r}_\phi$ as in (36).

2.11 Algorithmic summary: from $\mathcal{D}$ to $\mathbf{H}^2$

Given

$$\mathcal{D} = \{(\theta_i, x_i)\}_{i=1}^N, \tag{52}$$

the pipeline is:

1. Train $f_\phi(x, t, m, \Delta)$ with the pair-conditioned CTSM-v objective (30) on ordered sample pairs (as in the sampling scheme stated there).
2. For every ordered pair (i, j), compute m_{ij}, Δ_{ij}, and
$$\Delta L_{ij} = \int_0^1 \hat{s}_\phi(x_i, t, m_{ij}, \Delta_{ij})\, dt, \qquad \hat{s}_\phi = \mathbf{1}^\top f_\phi.$$
3. Form $H_{ij}^{\rightarrow} = 1 - \operatorname{sech}(\Delta L_{ij}/2)$ and $H_{ij}^{\mathrm{sym}} = \frac{1}{2}(H_{ij}^{\rightarrow} + H_{ji}^{\rightarrow})$.
4. Estimate $\mathbf{H}^2$ via (51) (e.g. binning or regression).

Thus

$$\{(\theta_i, x_i)\}_{i=1}^N \longrightarrow f_\phi \longrightarrow \Delta\mathbf{L} \longrightarrow \mathbf{H}^{\rightarrow} \longrightarrow \mathbf{H}^{\mathrm{sym}} \longrightarrow \mathbf{H}^2.$$

2.12 Evaluation protocol (preliminary)

For toy analyses, one may retain a fixed generative model (e.g. a low-dimensional linear tuning curve with heteroscedastic Gaussian noise), draw a large reference sample to approximate ground-truth H^2 and pairwise decoding matrices, then subsample the dataset repeatedly, re-estimate $\mathbf{H}^2$ and decoding via the pipeline above, and compare off-diagonal entries to the reference. Under this revised method, only the estimator for $\Delta\ell_{ij}(x)$ changes relative to a Fisher-score integration baseline; the downstream comparison protocol is unchanged.

3 PR-regularized autoencoder for low-to-high embedding

Many experiments require observations $x \in \mathbb{R}^d$ while a simple generative model lives in a low-dimensional latent space $\mathbb{R}^m$ with $m \ll d$. To generate such kind of dataset, we use *participation-ratio* (PR) regularized autoencoder to project low-dimensional dataset into high-dimensional dataset.

The PR-regularized autoencoder learns a deterministic map that lifts $z \in \mathbb{R}^m$ to embeddings $h \in \mathbb{R}^d$ without collapsing h onto a very thin subspace of $\mathbb{R}^d$. Fix $d \geq m$ and parameters ψ for an encoder $f_\psi : \mathbb{R}^m \to \mathbb{R}^d$ and decoder $g_\psi : \mathbb{R}^d \to \mathbb{R}^m$; write

$$h = f_\psi(z) \in \mathbb{R}^d, \qquad \hat{z} = g_\psi(h) \in \mathbb{R}^m. \tag{53}$$

Reconstruction is measured by mean squared error between $\hat{z}$ and z over training samples.

Given a mini-batch of embeddings $h^{(1)}, \ldots, h^{(n)}$, center them coordinatewise, let $\tilde{H} \in \mathbb{R}^{n \times d}$ have rows $(h^{(i)} - \bar{h})^\mathsf{T}$, and form the unbiased batch covariance $C = \frac{1}{n-1}\tilde{H}^\mathsf{T}\tilde{H}$. The *participation ratio* is

$$\mathrm{PR}(C) = \frac{\big(\operatorname{tr} C\big)^2}{\operatorname{tr}(C^2) + \varepsilon}, \tag{54}$$

with $\varepsilon > 0$ a small numerical floor; larger $\mathrm{PR}(C)$ indicates that batch variance spreads across more directions in $\mathbb{R}^d$. The training loss combines reconstruction and encouraging high participation,

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^{n} \big\| g_\psi(f_\psi(z^{(i)})) - z^{(i)} \big\|_2^2 - \lambda\, \mathrm{PR}(C), \tag{55}$$

with $\lambda \geq 0$ trading fidelity to z against spreading mass in the embedding space.

3.1 Path-integral θ-flow H-matrix agreement after PR embedding

After lifting low-dimensional observations into $\mathbb{R}^d$ with the PR-regularized autoencoder, we still wish to compare a *learned* pairwise information matrix to a *generative* reference. We summarize two regimes: embeddings from $m = 2$ to $d = 50$, and from $m = 10$ to $d = 100$. In each case the conditional law of x given θ follows the same random-amplitude Gaussian tuning family with observation noise scaled like $\sqrt{d}$ in the embedded space, so difficulty grows with d.

For nested training subsets of size n drawn from a larger reference pool, we train posterior and prior θ-space flow models and build the symmetric matrix H^{sym} using the *path-integral* construction (velocity-to-score along θ with trapezoidal integration), then bin to match the generative Monte Carlo estimate of (squared) Hellinger affinity between bin-conditional laws. We report the Pearson correlation between off-diagonal entries of the binned matrix and the corresponding square-root generative panel (top row of the convergence plots; panels annotate $\sqrt{H^2}$ vs. the learned summary). Separately, we fit pairwise linear classifiers between bins in observation space and compare their accuracy matrix to the one obtained at the reference sample size (bottom row).

Figure 3 collects the latent and embedded geometry (PCA views and tuning curves) together with these convergence panels. In both the $2 \to 50$ and $10 \to 100$ settings, the H-matrix track stays close to the generative ground truth even at $n = 80$: the blue correlation curve is already large and quickly plateaus near one. The pairwise-decoding track (red) is noisier at small n, as expected when separating high-dimensional clouds with limited data; it improves markedly by $n = 400$–600 and aligns with the H-matrix curve in the larger-n regime. The transient dip in decoding correlation at $n = 200$ in the $10 \to 100$ experiment is consistent with finite-sample variability of nested subsets and does not contradict strong agreement of the path-integral H summary with the generative reference. Overall, path-integral θ-flow estimation of the binned H structure remains a stable,

sample-efficient summary after PR embedding, while raw decoding metrics require more data to match the same reference.

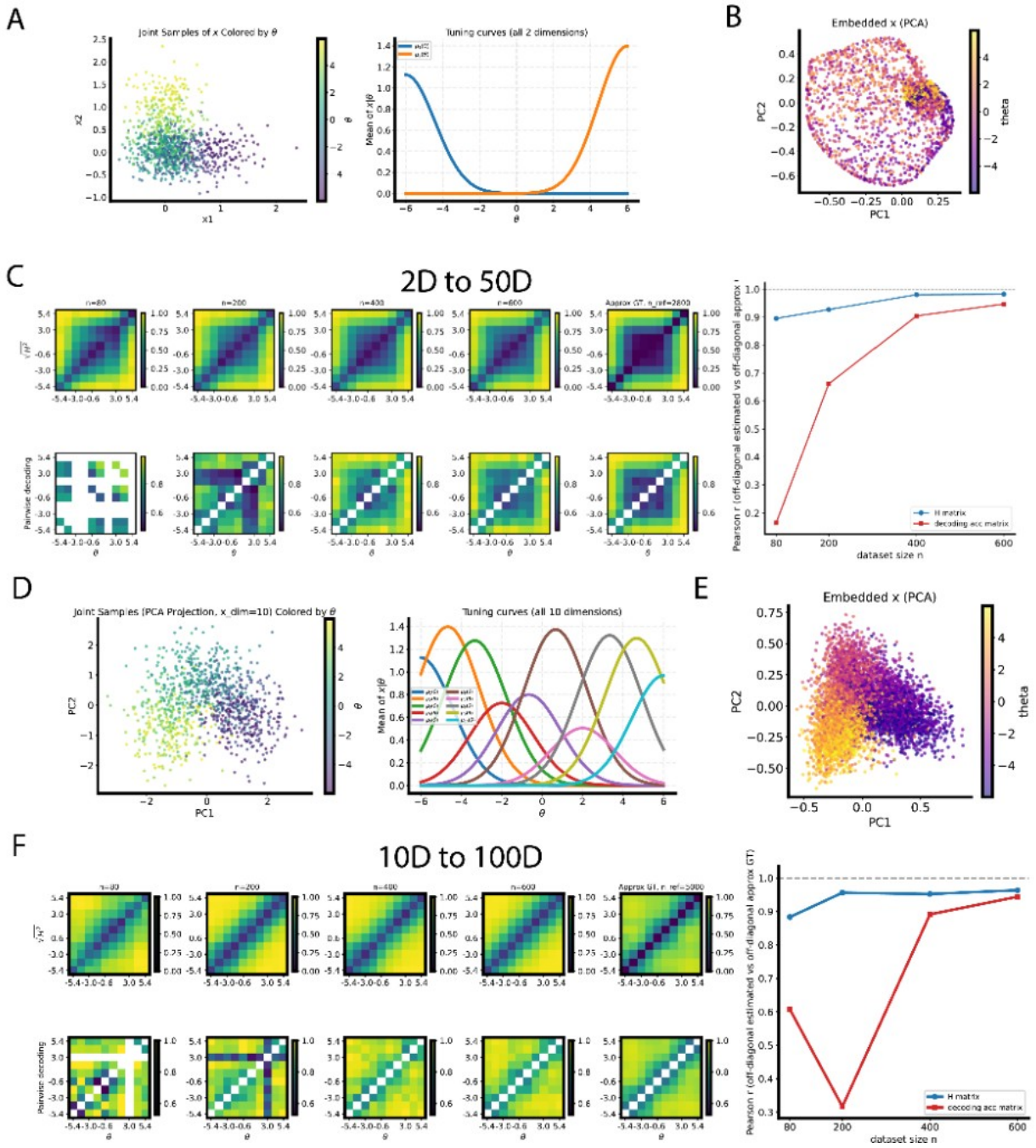

Figure 3: PR embedding of low-dimensional generative data into higher d, followed by path-integral θ-flow H-matrix and pairwise-decoding convergence versus sample size n. *Top:* $m = 2$ latent coordinates, tuning curves, PCA of embedded $x \in \mathbb{R}^{50}$, and binned $\sqrt{H^2}$ / decoding matrices for $n \in \{80, 200, 400, 600\}$ with a large-n_{ref} approximate ground-truth column; right: off-diagonal Pearson r vs n for the H track (blue) and decoding track (red). *Bottom:* the same layout for $m = 10$ and embedded $x \in \mathbb{R}^{100}$. The path-integral H correlation stays high even at small n; decoding correlation catches up as n grows.